DATE: 28 October 2022

DESIGNER: WLC

PROJECT No: WLC612-2

PROJECT NAME: A420 Signalised Junction

THE PROPOSALS HAVE BEEN DESIGNED TO MEET THE REQUIREMENTS OF THE FOLLOWING TARGET LIGHTING CLASS(ES):
C4 (SIGNALISED JUNCTION) IN ACCORDANCE WITH BS 5489:2020.

M4 - P2 COMPARATIVE (APPROACHES) IN ACCORDANCE WITH BS 5489:2020.

P3 (CYCLEPATH) IN ACCORDANCE WITH BS 5489:2020.

LEVELS ACHIEVED:
REFER TO INDIVIDUAL GRID WITHIN REPORT.

WLC612-2-LC-AC-001-R0

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Layout Report

General Data

Dimensions in Metres Angles in Degrees

Calculation Grids

ID	Grid Name	X	Y	X' Length	Y' Length	X' Spacing	Y' Spacing
1	Grid 1	420243.17	187039.44	375.00	322.35	1.50	1.50
2	Cyclepath	420423.32	187176.20	207.21	15.00	1.50	1.50
3	WEST APPROACH	420247.00	187109.93	117.00	54.00	1.50	1.50
4	EAST APPROACH	420454.79	187161.87	93.00	24.00	1.50	1.50
5	NORTH APPROACH	420259.68	187170.85	130.27	66.00	1.50	1.50

Luminaires

Luminaire A Data

Supplier	
Type	AXIA 3.2 5266 Integrated lenses - 24 OSLO N SQUARE GIANT@400m
Lamp(s)	24 OSLON SQUARE GIANT@400mA WW 730 230V 00-36-646
LampFlux(klm)/Colour	4.15 WW 3000K/70
File Name	AXIA 3.2 5266 24 OSLON SQUARE GIANT 400mA WW730 29.7W 430082 Integrated ...
Maintenance Factor	0.76
Lum. Int. Class	G2
No. in Project	3

Luminaire B Data

Supplier	Urbis Schreder
Type	AXIA 3.2 5279 Integrated lenses 24 OSLO N SQUARE GIANT@590mA
Lamp(s)	24 OSLON SQUARE GIANT@590mA WW 730 230V
LampFlux(klm)/Colour	5.67 WW 3000K/70
File Name	AXIA 3.2 5279 24 OSLON SQUARE GIANT 590mA WW 730 44W 430122 Integrated I...
Maintenance Factor	0.76
Lum. Int. Class	G2
No. in Project	17

Luminaire C Data

Supplier	Urbis Schreder
Type	AXIA 3.1 5266 Integrated lenses 16 OSLO N SQUARE GIANT@500mA
Lamp(s)	16 OSLON SQUARE GIANT@500mA WW 730 230V 01-37-043
LampFlux(klm)/Colour	3.29 WW 3000K/70
File Name	AXIA 3.1 5266 16 OSLON SQUARE GIANT 500mA WW 730 25.5W 429092 Integrated ...
Maintenance Factor	0.76
Lum. Int. Class	G1
No. in Project	6

Layout

ID	Type	X	Y	Height	Angle	Tilt	Cant	Out-reach	Target X	Target Y	Target Z
6	B	420266.62	187129.93	8.00	286.00	0.00	0.00	0.50			
7	B	420296.92	187141.13	8.00	286.00	0.00	0.00	0.50			
8	B	420301.22	187126.62	8.00	107.00	0.00	0.00	0.50			
10	B	420335.35	187135.48	8.00	108.00	0.00	0.00	0.50			
9	B	420330.83	187153.00	8.00	286.00	0.00	0.00	0.50			
11	B	420368.75	187144.15	8.00	102.00	0.00	0.00	0.50			

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Layout Continued

ID	Type	X	Y	Height	Angle	Tilt	Cant	Out-reach	Target X	Target Y	Target Z
12	B	420364.50	187162.52	8.00	286.00	0.00	0.00	0.50			
13	B	420400.44	187152.87	8.00	106.00	0.00	0.00	0.50			
14	B	420402.91	187174.88	8.00	286.00	0.00	0.00	0.50			
4	B	420388.26	187178.56	8.00	212.00	0.00	0.00	0.50			
3	A	420348.98	187192.40	5.00	251.00	0.00	0.00	0.50			
15	B	420434.89	187160.71	8.00	103.00	0.00	0.00	0.50			
16	B	420443.55	187185.36	8.00	286.00	0.00	0.00	0.50			
17	B	420464.39	187169.62	8.00	110.00	0.00	0.00	0.50			
18	B	420476.33	187190.86	8.00	286.00	0.00	0.00	0.50			
19	B	420495.08	187179.34	8.00	118.00	0.00	0.00	0.50			
20	B	420527.06	187190.37	8.00	110.00	0.00	0.00	0.50			
2	A	420322.19	187203.56	5.00	239.00	0.00	0.00	0.40			
1	A	420290.47	187219.47	5.00	246.00	0.00	0.00	0.40			
5	B	420370.18	187173.26	8.00	17.00	0.00	0.00	0.50			
1F	C	420426.00	187185.30	6.00	286.00	0.00	0.00	0.40			
2F	C	420450.76	187194.52	6.00	292.00	0.00	0.00	0.40			
3F	C	420493.29	187210.61	6.00	297.00	0.00	0.00	0.40			
4F	C	420533.31	187224.81	6.00	295.00	0.00	0.00	0.40			
5F	C	420573.01	187239.23	6.00	293.00	0.00	0.00	0.40			
6F	C	420599.65	187249.38	6.00	312.00	0.00	0.00	0.40			

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Horizontal Illuminance (lux)

Grid 1

Results

Eav	10.87
Emin	4.33
Emax	24.17
Emin/Emax	0.18
Emin/Eav	0.40

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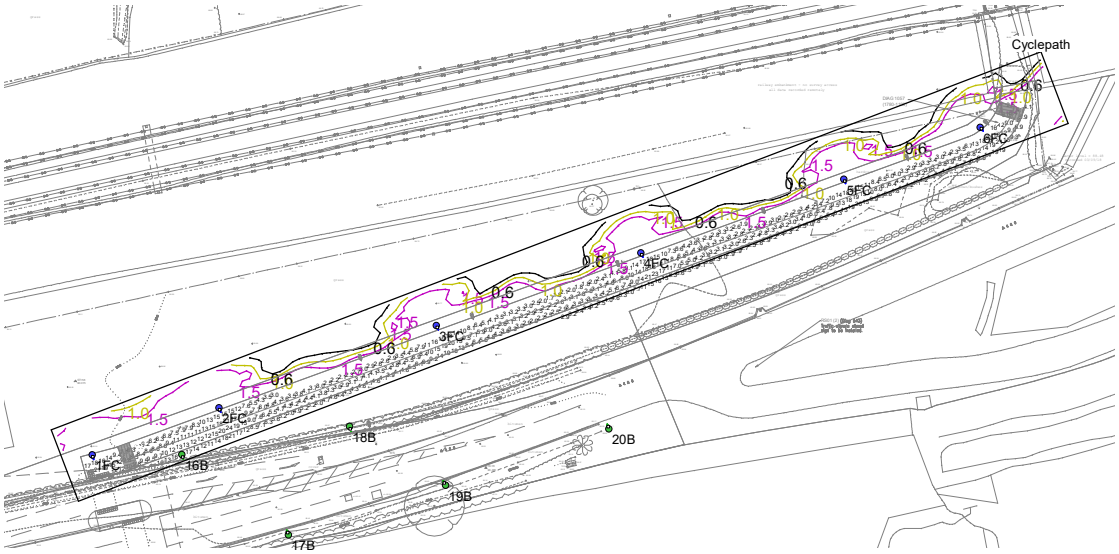
Horizontal Illuminance (lux)

Cyclepath



Results

Eav	7.63
Emin	1.72
Emax	26.43
Emin/Emax	0.07
Emin/Eav	0.23

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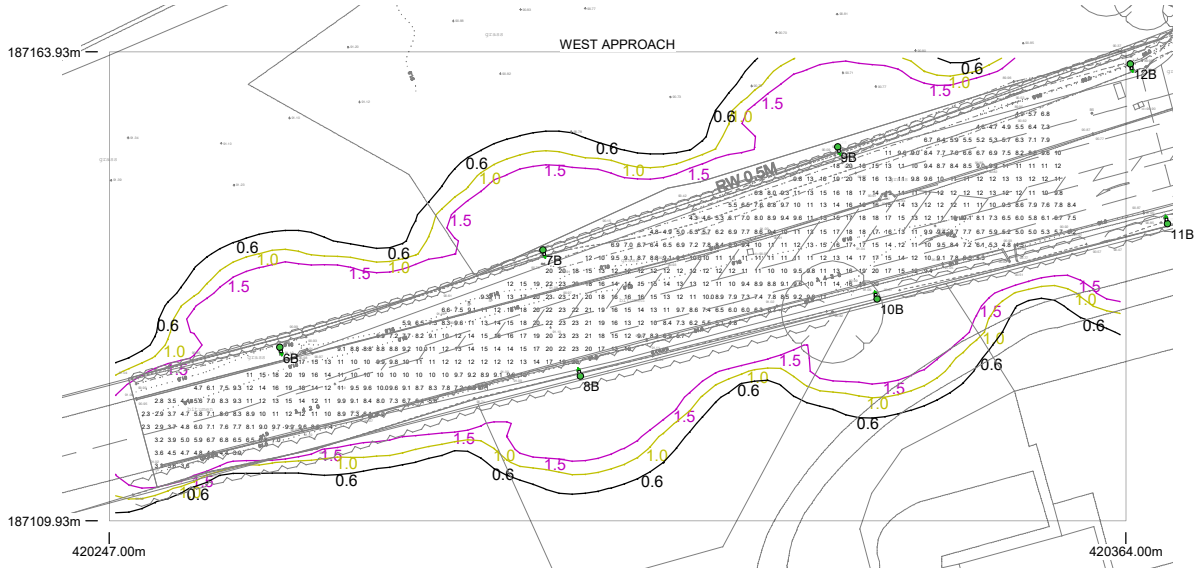
Horizontal Illuminance (lux)

WEST APPROACH



Results

Eav	10.98
Emin	2.31
Emax	23.46
Emin/Emax	0.10
Emin/Eav	0.21

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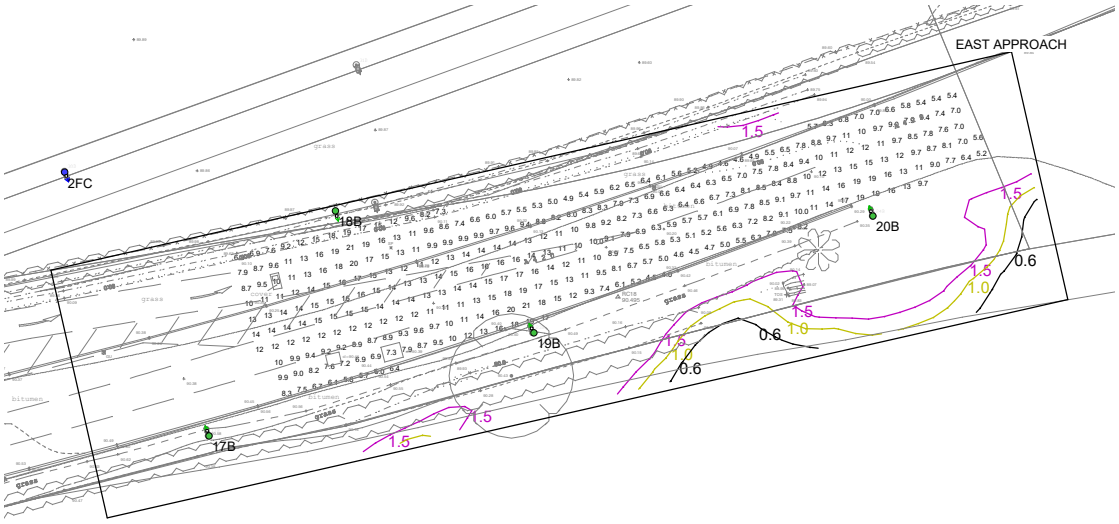
Horizontal Illuminance (lux)

EAST APPROACH



Results

Eav	10.45
Emin	3.98
Emax	20.64
Emin/Emax	0.19
Emin/Eav	0.38

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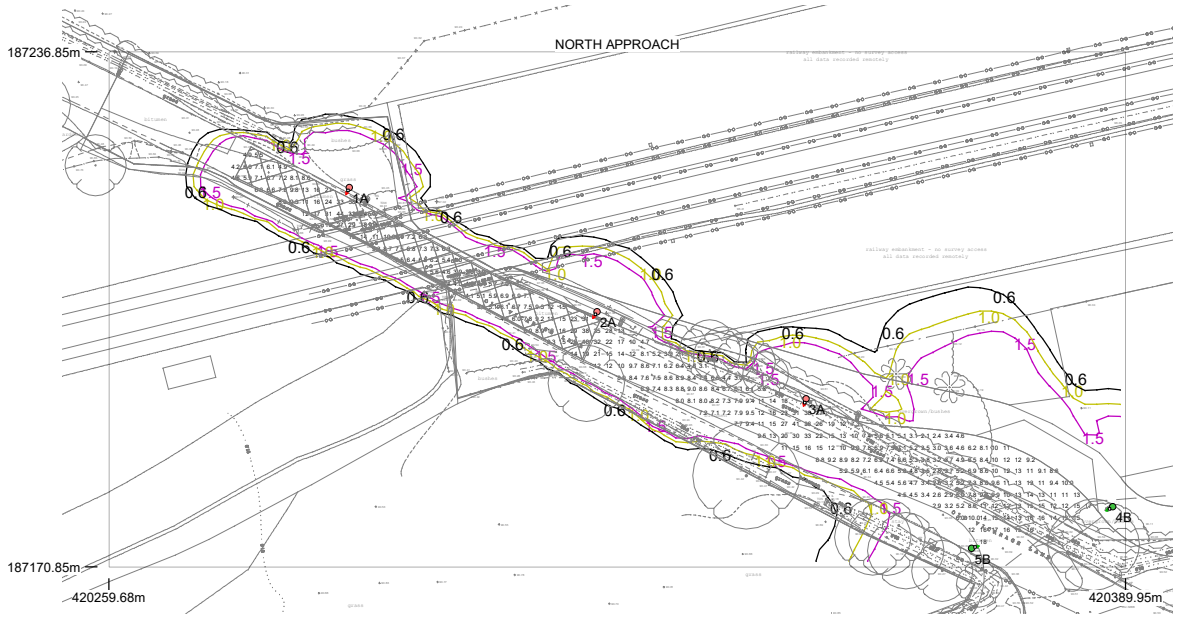
Horizontal Illuminance (lux)

NORTH APPROACH



Results

Eav	10.87
Emin	2.07
Emax	43.84
Emin/Emax	0.05
Emin/Eav	0.19

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